

Definition 17.1

A *ring* is set R equipped with two binary operations:

- *addition*, denoted $a + b$
- *multiplication*, denoted $a \cdot b$

satisfying the following properties:

- 1) R taken with addition is an abelian group (with the identity element $0 \in R$).
- 2) Multiplication is associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for any $a, b, c \in R$.
- 3) For any $a, b, c \in R$ we have $(a + b)c = ac + bc$ and $(b + c)a = ba + ca$.

Definition 17.2

We say that a ring R is *commutative* if $ab = ba$ for any $a, b \in R$.

We say that R is a *ring with unity* if there is an element $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$.

Theorem 17.3

If a ring R has a unity, the the unity is unique.

Theorem 17.4

If a ring R . For any $a, b \in R$ we have:

1) $0 \cdot a = a \cdot 0 = 0$.

2) $a(-b) = (-a)b = -(ab)$.

3) $(-a)(-b) = ab$

4) if R has a unity $1 \in R$ then $(-1)a = a(-1) = -a$.

Definition 17.5

Let R be a ring. A *subring* of R is a subset $S \subseteq R$ such that S is a ring with respect to the addition and multiplication in R .

Theorem 17.6

Let R be a ring. A subset $S \subseteq R$ is a subring of R if and only if the following conditions are satisfied:

- 1) $0 \in S$
- 2) if $a, b \in S$ then $a + b \in S$ and $ab \in S$
- 3) if $a \in S$ then $(-a) \in S$

Definition 17.7

The *direct product* of rings R_1, R_2 is a ring $R_1 \times R_2$ defined as follows:

- Elements of $R_1 \times R_2$ are ordered tuples (a_1, a_2) where $a_i \in R_i$
- Addition and multiplication are given by

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$$

$$(a_1, a_2) \cdot (b_1, b_2) = (a_1 b_1, a_2 b_2)$$